databricks

Introduction to Data Engineering in Databricks

LECTURE

Data Engineering in Databricks

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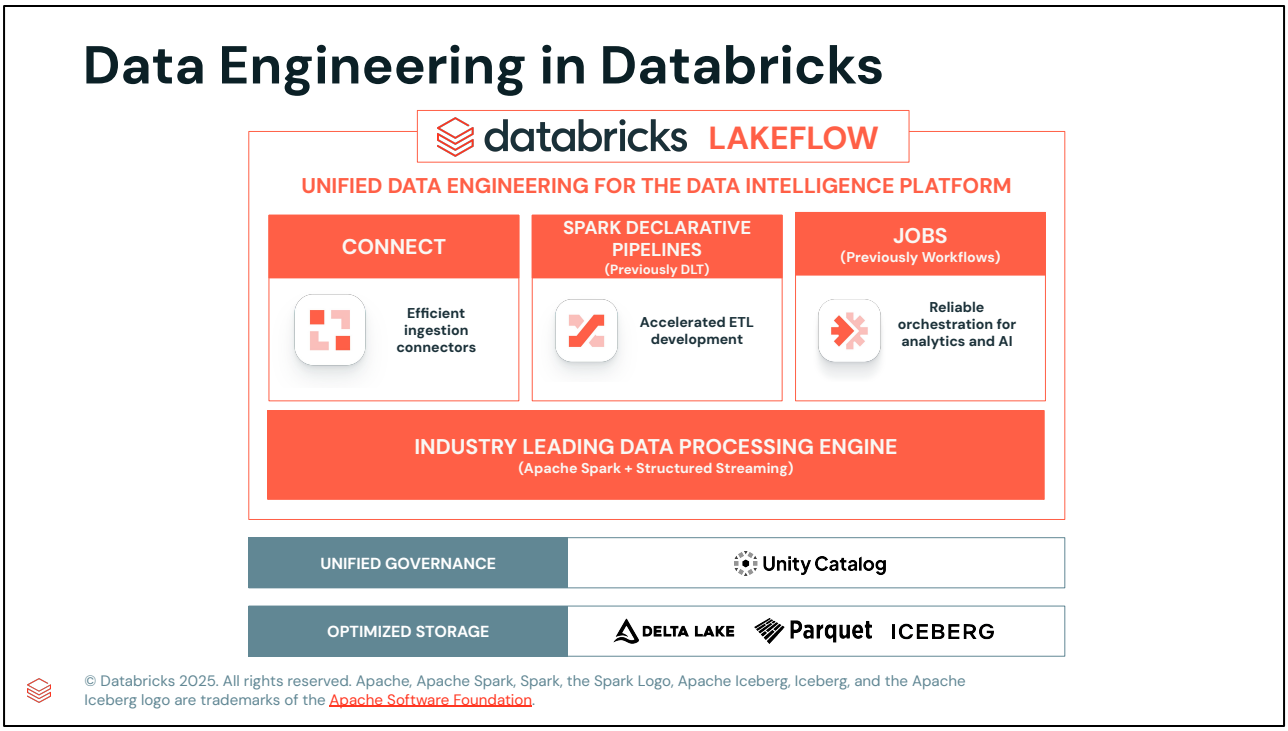
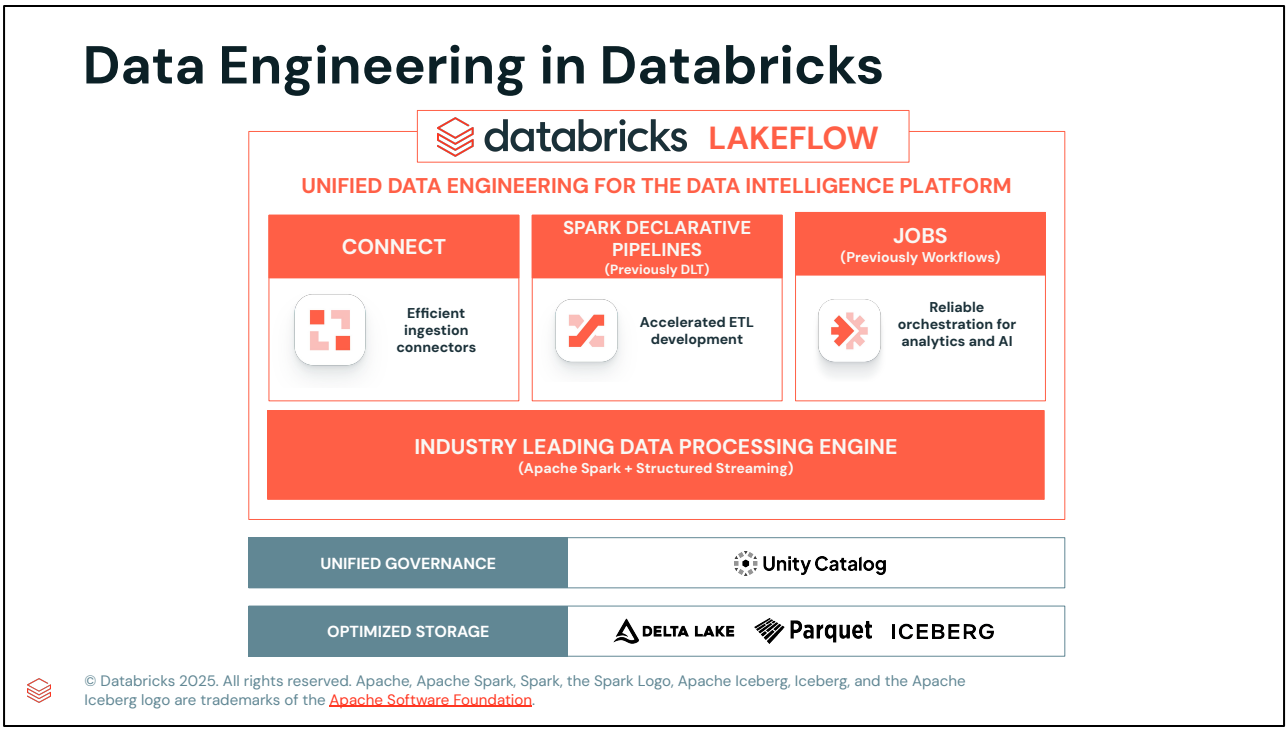
In this lecture, you will learn how Databricks enables efficient data engineering through optimized storage, unified data governance with Unity Catalog, and Lakeflow's integrated tools for data ingestion, transformation, and orchestration—focusing on simplifying and automating reliable batch and streaming ETL with Lakeflow Declarative Pipelines.



It all begins with optimized storage using Delta Lake, Parquet, or Iceberg.

Built on top of this storage layer is unified governance with Unity Catalog. Unity Catalog is a centralized data catalog that provides access control, auditing, data lineage, quality monitoring, and data discovery across Databricks workspaces.

Databricks then offers **Lakeflow**, an end-to-end data engineering solution that empowers data engineers, software developers, SQL developers, analysts, and data scientists to deliver high-quality data for downstream analytics, AI, and operational applications. Lakeflow provides a unified platform for data ingestion, transformation, and orchestration, and includes the following components:



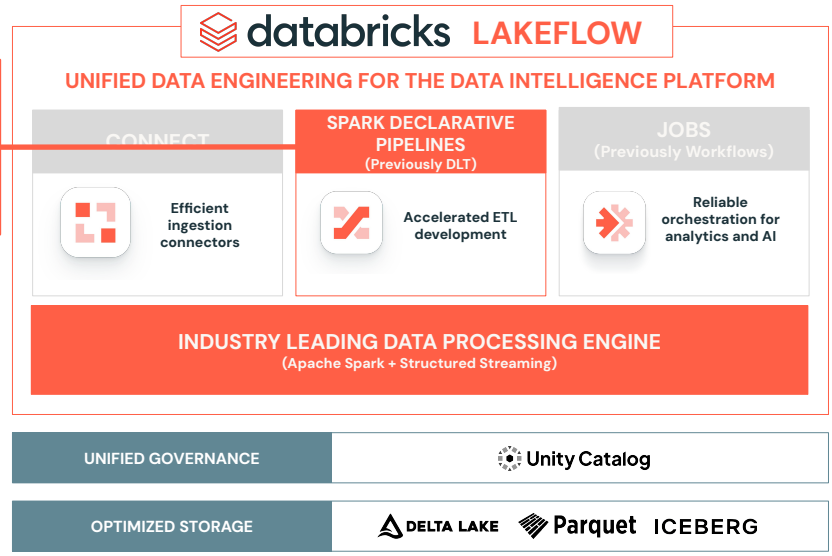
- **Lakeflow Connect**: A set of efficient ingestion connectors that simplify data ingestion from popular enterprise applications, databases, cloud storage, message buses, and local files.

- **Lakeflow Spark Declarative Pipelines**: A framework for building batch and streaming data pipelines using SQL and Python, designed to accelerate ETL development.

- **Lakeflow Jobs**: A workflow automation tool for Databricks that orchestrates data processing workloads. It enables coordination of multiple tasks within complex workflows, allowing for the scheduling, optimization, and management of repeatable processes.

Data Engineering in Databricks

Simplifying batch and streaming ETL with automated reliability, optimizations and built-in data quality with **Lakeflow Spark Declarative Pipelines (SDP)**!



UNIFIED GOVERNANCE	Unity Catalog
OPTIMIZED STORAGE	DELTA LAKE Parquet ICEBERG

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This course will focus on simplifying batch and streaming ETL with automated reliability, optimizations and built-in data quality with Lakeflow Spark Declarative Pipelines!

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Thank you for completing this lesson and continuing your journey to develop your skills with us.